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Project DOCUMENTATION

(Portfolio and Management System)

for

Aluminum and Glass Fabrication Company

By

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1. Introduction

1.1 Problem Statement

Traditional management of aluminum and glass manufacturing relies heavily on manual processes, spreadsheets, and disconnected systems, leading to inefficiencies, delays, and errors. Lack of integration between procurement, production, inventory, logistics, and customer service causes poor traceability, resource wastage, and weak decision-making.

1.2 Purpose

The purpose of this system is to provide a centralized web application that manages the complete workflow of an aluminum and glass manufacturing company. To stay competitive, deliver consistent quality streamline workflows and ensure transparency across all operations.

1.3 Goals and Objectives

The goal of this system is to transform aluminum and glass manufacturing operations into a fully digital, efficient, and customer-focused process. To achieve this, the system will:

- **Automate and integrate workflows** across procurement, production, inventory, logistics, installation, and customer service.
- **Ensure accuracy and transparency** through real-time tracking, reporting, and role-based access.
- **Optimize resource utilization** by minimizing waste, improving scheduling, and enhancing productivity.
- **Improve decision-making** with dashboards, analytics, and project profitability insights.

2. Roles in the System

1. **Admin**
    - Full control of the system, user management, role assignment, and global settings.
2. **Procurement Manager**
    - Handles raw material purchase (aluminum profiles, glass sheets, hardware, adhesives, seals, etc.).
    - Vendor negotiations, purchase orders, supplier database.
3. **Inventory/Warehouse Manager**
    - Tracks stock of raw materials, semi-finished goods, and finished products.
    - Issues materials to production.

4. **Production/Factory Manager**
    - Plans and schedules production orders.
    - Assigns machines/workers for tasks (cutting, tempering, paint coating, assembling).
    - Tracks work progress and rejects.
5. **Quality Control Officer**
    - Inspects products at different stages (glass cutting, aluminum finishing, final assembly).
    - Logs defects and approves/rejects batches.
6. **Project Manager**
    - Oversees customer projects (commercial/residential).
    - Tracks progress from design → production → delivery → installation.
    - Assigns field teams.
7. **Sales & Marketing**
    - Handles customer inquiries, quotations, orders.
    - CRM (customer relationship management).
    - Tracks conversion from leads to orders.
8. **Finance/Accounts**
    - Manages invoices, supplier payments, customer billing.
    - Tracks project profitability.
9. **Logistics/Delivery Manager**
    - Plans delivery schedules for finished goods.
    - Manages transport fleet.
    - Tracks shipping and delivery confirmation.
10. **Installation Team Lead**
    - Receives job orders from project manager.
    - Schedules installation teams.
    - Logs installation completion and customer sign-off.

3. Functional Requirements (by Module)

3.1 Procurement & Supplier Management

Manage all supplier-related activities efficiently.

- **Supplier Directory:** Store contacts, lead times, payment terms. Track supplier ratings & history.
- Request for Quotation → Supplier Quotes → Supplier selection.
- Purchase Order creation & approval workflow.
- Receive Goods and Auto-update inventory.
- Supplier invoices matching, returns, and debit notes.
- Automatic Re-order Alerts when stock reaches Minimum levels.

3.2 Inventory & Warehouse

- **Material:** aluminum profiles (by profile type & alloy), glass types (float, tempered, laminated), hardware, consumables.
- Batch and lot tracking.
- Locations: multi-warehouses
- Stock Movements Tracking: Goods Receipts, stock issue, stock transfer, adjustment.

3.3 Design & Engineering

- Parametric inputs (width, height, mullion spacing, glass thickness).
- Bill of material generation (parts list with quantities, profile cuts, glass sizes).
- Cutting list export (for CNC / cutting machines).
- Approvals: Customer & internal sign-off.

3.4 Production Management

- Production orders created from sales / project.
- **Work routing:** cut → machine operations (milling/pressing) → surface treatment (paint job) → assembly → packing.
- Job scheduling with resource (machine & human operator) allocation.
- Machine & shift scheduling, maintenance history.
- Work in progress tracking and real-time status (Pending / In process / Completed).
- Scrap & status logging.

3.5 Quality Control (QC)

- QC checks at different stages (incoming, midline, final).
- **Checklists for Quality:** dimensional checks, finish quality, glass integrity, hardware fit.
- Defectlogging (in case some pieces damage at final stages) and corrective action assignment.

3.6 Sales & CRM

- Lead (Customer) capture & pipeline stages.
- Quotation builder for rough estimate of final order: select product type, dimensions, materials, add markup, taxes.
- Price catalogs and discount rules.
- Order-to-project conversion by customer approvals.

3.7 Project Management & Scheduling

- Project creation (site, contact, delivery/installation dates).
- Milestones (design approval, production start, delivery, installation).
- Site-specific documents: site measurements, photos, access notes.

3.8 Logistics & Delivery

- Create Delivery Challan / Packing List.
- Fleet management: assign vehicle & driver, delivery dates, route assignments.
- POD (Proof of Delivery) with e-signature or photo upload.

3.9 Installation Management

- Installation of work orders on site.
- Schedule installers, list required items per job.
- Final Installation Proof by final photos.

3.10 Finance & Accounting

- Sales invoicing, credit notes, payment receipts.
- Supplier bills and payments.
- Project costing with **General Ledger** mapping.
- Tax calculation, VAT/GST handling.

3.11 Reporting & Analytics

- **Standard reports:** stock aging, material consumption, production efficiency, job cycle time, scrap rate, on-time delivery, profitability per project.
- Custom report builder and CSV/Excel export.
- Scheduled reports & email distribution.

4. Non-Functional Requirements (NFRs)

4.1 Performance

- The system should run **smoothly under normal and peak workloads**.
- It must provide **fast and reliable responses** for all key operations.

4.2 Scalability

- The system should be **expandable to support multiple factories and branches**.
- It must handle growth in data and users without affecting performance.

4.3 Security

- **Role-based access control** for different user levels.
- **Encryption** of sensitive information such as passwords and financial data.
- Regular **backups and recovery mechanisms** to prevent data loss.

4.4 Usability

- User interface should be **simple, intuitive, and mobile-friendly**.
- Easy to navigate for both technical and non-technical staff.
- Provide **multi-language support** if required.

5. Database Schema (Entities)

5.1 Users & Roles

- **Employees** (employee_id, first_name, last_name, email, phone, address, hire_date, status, baseSalary)
- **EmployeeAttendance** (attendance_id, employee_id, date, check_in_time, Check_out_time, status [Present / Absent / Late / Half-day], totalHours)
- **AbsenceTracking** (absence_id, employee_id, start_date, end_date, reason, approved_by, status [Pending / Approved / Rejected/ Uninformed])
- **Roles** (role_id, role_name, description)
- **Accounts** (account_id, employee_id, username, password_hash, role_id, is_active, last_login)
- **Bonus Tracking** (bonus_id, employee_id, amount, reason, awarded_by, status[]Pending / Approved / Paid)

Relationships:

- Employees. employee_id → Accounts. employee_id (M:1)
- Role.role_id → Accounts.role_id
- EmployeeAttendance.employee_id → Employees.employee_id (M:1)
- EmployeeBonus.employee_id → Employees.employee_id (M:1)
- EmployeeBonus.awarded_by → Employees.employee_id (M:1)

5.2 Suppliers & Customer

- **Suppliers** (supplier_id, name, contact_info, rating, email, address, payment_terms)
- **Customers** (customer_id, name, phone, email, address, city, type [retail/contract])

5.3 Inventory & Warehouse

- **RawMaterials** (material_id, type_name **[**Aluminium, Glass, Screw, Glue, Rubber, Tools**]**, category, unit, size, thickness, colour, current_stock, reorder_level, supplier_id)
- **InventoryBatches** (batch_id, material_id, warehouse_id, qty_available, received_date)
- **Warehouses** (warehouse_id, name, location)
- **StockMovements** (movement_id, batch_id, from_where, to_where, qty, movement_type, created_at)

Relationships:

- **InventoryBatches**.material_id → **RawMaterials.material_**id (M:1)
- **InventoryBatches**.warehouse_id → **Warehouses**.warehouse_id (M:1)
- **StockMovements.**batch_id → **InventoryBatches**.batch_id (M:1)

5.4 Sales & Quotations

- **Quotations** (quotation_id, customer_id, created_by, date, status [pending/approved/rejected], total_price, notes)
- **QuotationItems** (item_id, quotation_id, material_id, description, qty, unit_price, total)
- **SalesOrders** (order_id, quotation_id, customer_id, created_by, order_date, status [in-progress/completed/cancelled])
- **OrderItems** (order_item_id, order_id, material_id, description, qty, unit_price, total)

Relationships:

- QuotationItems.quotation_id → Quotations.quotation_id (M:1)
- SalesOrders.quotation_id → Quotations.quotation_id (M:1)
- SalesOrders.customer_id → Customers.customer_id (M:1)
- OrderItems.order_id → SalesOrders.order_id (M:1)
- OrderItems.material_id → RawMaterials.material_id (M:1) QuotationItems.material_id → RawMaterials.material_id (M:1)

5.5 Production & Work Orders

- **ProductionStages** (stage_id, name [Measurement, Cutting, Colouring, Assembly, QC, Delivery, Installation], description)
- **ProductionTasks** (task_id, order_id, stage_id, assigned_to, supervisor_id, status [assigned/completed/pending], start_date, end_date, remarks)
- **TaskLogs** (log_id, task_id, action_by, action_type [assigned, started, completed], date_time, comments)

Relationships:

- ProductionTasks.order_id → SalesOrders.order_id (M:1)
- ProductionTasks.stage_id → ProductionStages.stage_id (M:1)
- ProductionTasks.assigned_to → Employees.employee_id (M:1) *(assuming a table for employees)*
- ProductionTasks.supervisor_id → Employees.employee_id (M:1)
- TaskLogs.task_id → ProductionTasks.task_id (M:1)
- TaskLogs.action_by → Employees.employee_id (M:1)

5.6 Quality Control

- **QCInspections** (qc_id, task_id, inspector_id, date, result, remarks)
- **Defects** (defect_id, qc_id, defect_type, description, severity, corrective_action)

Relationship:

- QCInspections.task_id → ProductionTasks.task_id (M:1)
- Defects.qc_id → QCInspections.qc_id (M:1)

5.7 Logistics & Delivery

- **Delivery** (delivery_id, order_id, driver_id, vehicle_no, scheduled_date, actual_date, status)

- **DeliveryItems** (delivery_item_id, delivery_id, order_item_id, qty, remarks)
- **Vehicles** (vehicle_id, vehicle_no, capacity)

Relationships:

- Delivery.order_id → SalesOrders.order_id (M:1)
- Delivery.driver_id → Employees.employee_id (M:1)
- DeliveryItems.delivery_id → Delivery.delivery_id (M:1)
- DeliveryItems.order_item_id → OrderItems.order_item_id (M:1)
- Delivery.vehicle_no → Vehicles.vehicle_no (M:1)

5.8 Installations

- **Installation** (install_id, order_id, team_lead_id, scheduled_date, actual_date, status, remarks)
- **InstallationReports** (report_id, install_id, notes, customer_signature, photos)

Relationships:

- Installation.order_id → SalesOrders.order_id (M:1)
- Installation.team_lead_id → Employees.employee_id (M:1)
- InstallationReports.install_id → Installation.install_id (M:1)

5.9 Finance

- **Invoices** (invoice_id, order_id, customer_id, issue_date, due_date, total_amount, status [paid/unpaid/partial])
- **InvoiceItems** (invoice_item_id, invoice_id, material_id, description, qty, unit_price, total)
- **PurchaseInvoices** (purchase_invoice_id, supplier_id, date, total_amount, status [paid/unpaid])
- **PurchaseInvoiceItems** (purchase_item_id, purchase_invoice_id, material_id, qty, unit_price, total)
- **InventoryTransactions** (transaction_id, material_id, qty, transaction_type [in/out], date, linked_task_id, purchase_invoice_id)
- **Payments** (payment_id, invoice_id, amount, method [cash/bank/cheque/online], date, received_by)
- **Expenses** (expense_id, category, description, amount, date, added_by)

6. Database Workflow (How Data Flows)

Step 1: Order Workflow

Quotation → Sales Order

Sales staff creates Quotation → Customer accepts → Becomes SalesOrder.

OrderItems reserved from RawMaterials stock.

Step 2: Production Workflow

Supervisor assigns ProductionTasks (Cutting, Colouring, Assembly, etc.).

Workers update status → logged in TaskLogs.

Materials consumed → recorded in InventoryTransactions.

Step 3: Delivery & Installation

Once order is complete → linked to Delivery & Installation.

Team assigned, delivery scheduled, customer signs off.

Step 4: Billing & Accounting

System generates Invoice from SalesOrder.

Customer pays → recorded in Payments.

Raw materials purchased → recorded in PurchaseInvoices & PurchaseInvoiceItems.

Expenses (transport, labour) logged in Expenses.

Step 5: Reporting & Decision Making

InventoryTransactions + Invoices = Profitability reports.

ProductionTasks + TaskLogs = Employee performance.

Customers + SalesOrders = Sales trends.

Suppliers + PurchaseInvoices = Supplier performance.

7. Project Workflow Diagram